DBMS - SQL Query Implementation for E-Commerce Database

Lenskart E-Commerce Order Management Database

1. SELECT, WHERE, ORDER BY, and DISTINCT Queries

1.1 Display All Products

```
SQL> SELECT * FROM LenskartProduct;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         1
Classic Black Frame
Vincent Chase
      1200 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         2
Blue Light Glasses
John Jacobs
      1800 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         3
Round Metal Frame
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         4
Classic Black Sunglasses
Fastrack
      2000 Sunglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         5
Daily Contact Lenses
Bausch
       900 Contact Lenses
```

1.2 Search Products by Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM LenskartProduct
  3  WHERE Price > 1500;
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         2
Blue Light Glasses
      1800

         4
Classic Black Sunglasses
      2000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

1.3 Sort Products by Price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM LenskartProduct
  3  ORDER BY Price ASC;
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         5
Daily Contact Lenses
       900

         1
Classic Black Frame
      1200

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

         3
Round Metal Frame
      1500

         2
Blue Light Glasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1800

         4
Classic Black Sunglasses
      2000
```

1.4 Display Unique Categories

```
SQL> SELECT DISTINCT Category
  2  FROM LenskartProduct;
```

```
CATEGORY
--------------------------------------------------
Eyeglasses
Sunglasses
Contact Lenses
```

1.5 Search Products by Category

```
SQL> SELECT Product_ID, Product_Name, Brand, Price
  2  FROM LenskartProduct
  3  WHERE Category = 'Eyeglasses';
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE
----------
         1
Classic Black Frame
Vincent Chase
      1200


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE
----------
         2
Blue Light Glasses
John Jacobs
      1800


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE
----------
         3
Round Metal Frame
Lenskart Air
      1500
```

2. Search Products Based on Price, Category, and Availability

2.1 Products Priced Below 1500

```
SQL> SELECT * FROM LenskartProduct
  2  WHERE Price < 1500;

PRODUCT_ID PRODUCT_NAME
---------- --------------------------------------------------------------------------------------------------------------
BRAND                                                                                                             PRICE
-------------------------------------------------------------------------------------------------------------- ----------
CATEGORY
--------------------------------------------------
         1 Classic Black Frame
Vincent Chase                                                                                                       1200
Eyeglasses

         5 Daily Contact Lenses
Bausch                                                                                                               900
Contact Lenses
```

2.2 Products Priced Between 1000 and 2000

```
SQL> SELECT * FROM LenskartProduct
  2  WHERE Price BETWEEN 1000 AND 2000;
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         1
Classic Black Frame
Vincent Chase
      1200 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         2
Blue Light Glasses
John Jacobs
      1800 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         3
Round Metal Frame
Lenskart Air
      1500 Eyeglasses
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         4
Classic Black Sunglasses
Fastrack
      2000 Sunglasses
```

2.3 Search Products by Category

```
SQL> SELECT * FROM LenskartProduct
  2  WHERE Category = 'Eyeglasses';
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         1
Classic Black Frame
Vincent Chase
      1200 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         2
Blue Light Glasses
John Jacobs
      1800 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         3
Round Metal Frame
Lenskart Air
      1500 Eyeglasses
```

2.4 Display Available Products

```
SQL> SELECT P.Product_ID, P.Product_Name, I.Stock_Quantity, I.Stock_Status
  2  FROM LenskartProduct P
  3  JOIN LenskartInventory I
  4  ON P.Product_ID = I.Product_ID
  5  WHERE I.Stock_Status = 'Available';
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
         1
Classic Black Frame
            50 Available

         2
Blue Light Glasses
            30 Available

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------

         3
Round Metal Frame
            25 Available

         4
Classic Black Sunglasses

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
            40 Available
```

2.5 Display Unavailable Products

```
SQL> SELECT P.Product_ID, P.Product_Name, I.Stock_Quantity, I.Stock_Status
  2  FROM LenskartProduct P
  3  JOIN LenskartInventory I
  4  ON P.Product_ID = I.Product_ID
  5  WHERE I.Stock_Status = 'Unavailable';
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
STOCK_QUANTITY STOCK_STATUS
-------------- --------------------
         5
Daily Contact Lenses
             0 Unavailable
```

3.1 Display Customer Names and Email Addresses

```
SQL> SELECT Customer_Name, Email
  2  FROM LenskartCustomer;
```

```
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
Arun Kumar
arun@gmail.com

Priya Sharma
priya@gmail.com

Rahul Kumar
rahul@gmail.com


CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
Divya Raj
divya@gmail.com

Karthik S
karthik@gmail.com
```

3.2 Retrieve Customers from Chennai

```
SQL> SELECT *
  2  FROM LenskartCustomer
  3  WHERE Address = 'Chennai';
```

```
CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
ADDRESS
--------------------------------------------------------------------------------
         1
Arun Kumar
arun@gmail.com

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
ADDRESS
--------------------------------------------------------------------------------
9876500001
Chennai
```

3.3 Retrieve Orders with Customer Information

```
SQL> SELECT O.Order_ID, C.Customer_Name, O.Order_Date,
  2         O.Total_Amount, O.Order_Status
  3  FROM LenskartOrder O
  4  JOIN LenskartCustomer C
  5  ON O.Customer_ID = C.Customer_ID;
```

```
  ORDER_ID
----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ ------------------------------
         1
Arun Kumar
18-SEP-26         1200 Confirmed

         2
Priya Sharma
19-SEP-26         1800 Shipped

  ORDER_ID
----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ ------------------------------

         3
Rahul Kumar
20-SEP-26         1500 Delivered

         4
Divya Raj

  ORDER_ID
----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------------ ------------------------------
21-SEP-26         2000 Confirmed

         5
Karthik S
22-SEP-26          900 Processing
```

3.4 Retrieve Products with Seller Information

```
SQL> SELECT P.Product_Name, S.Seller_Name, S.Address
  2  FROM LenskartProduct P
  3  JOIN LenskartInventory I
  4  ON P.Product_ID = I.Product_ID
  5  JOIN LenskartSeller S
  6  ON I.Seller_ID = S.Seller_ID;
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
SELLER_NAME
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
Classic Black Frame
Vision Plus
Chennai

Blue Light Glasses
Eye World
Bangalore

PRODUCT_NAME
--------------------------------------------------------------------------------
SELLER_NAME
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------

Round Metal Frame
Optic Zone
Coimbatore

Classic Black Sunglasses
Clear Vision Store

PRODUCT_NAME
--------------------------------------------------------------------------------
SELLER_NAME
--------------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------------
Madurai

Daily Contact Lenses
Lens World
Salem
```

4. Apply Filtering Conditions

4.1 Filter Orders by Amount

```
SQL> SELECT *
  2  FROM LenskartOrder
  3  WHERE Total_Amount > 2000;
```

```
no rows selected
```

4.2 Filter Orders by Status

```
SQL> SELECT *
  2  FROM LenskartOrder
  3  WHERE Order_Status = 'Confirmed';
```

```
  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ ------------------------------
         1           1 18-SEP-26         1200 Confirmed
         4           4 21-SEP-26         2000 Confirmed
```

4.3 Search Customers by Name

```
SQL> SELECT *
  2  FROM LenskartCustomer
  3  WHERE Customer_Name LIKE 'A%';
```

```
CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
ADDRESS
--------------------------------------------------------------------------------
         1
Arun Kumar
arun@gmail.com

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
PHONE_NUMBER
---------------
ADDRESS
--------------------------------------------------------------------------------
9876500001
Chennai
```

4.4 Filter Products by Price Range

```
SQL> SELECT *
  2  FROM LenskartProduct
  3  WHERE Price BETWEEN 1000 AND 1800;
```

```
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         1
Classic Black Frame
Vincent Chase
      1200 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         2
Blue Light Glasses
John Jacobs
      1800 Eyeglasses


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND
--------------------------------------------------------------------------------
     PRICE CATEGORY
---------- --------------------------------------------------
         3
Round Metal Frame
Lenskart Air
      1500 Eyeglasses
```

4.5 Display Orders That Are Not Delivered

```
SQL> SELECT *
  2  FROM LenskartOrder
  3  WHERE Order_Status <> 'Delivered';
```

ORDER_ID	CUSTOMER_ID	ORDER_DAT	TOTAL_AMOUNT	ORDER_STATUS
1	1	18-SEP-26	1200	Confirmed
2	2	19-SEP-26	1800	Shipped
4	4	21-SEP-26	2000	Confirmed
5	5	22-SEP-26	900	Processing

5. Generate Basic Business Reports

5.1 Total Number of Products

```
SQL> SELECT COUNT(*) AS Total_Products
  2  FROM LenskartProduct;

TOTAL_PRODUCTS
--------------
             5
```

5.2 Total Number of Customers

```
SQL> SELECT COUNT(*) AS Total_Customers
  2  FROM LenskartCustomer;

TOTAL_CUSTOMERS
---------------
              5
```

5.3 Total Number of Orders

```
SQL> SELECT COUNT(*) AS Total_Orders
  2  FROM LenskartOrder;

TOTAL_ORDERS
------------
           5
```

5.4 Calculate Total Sales Amount

```
SQL> SELECT SUM(Total_Amount) AS Total_Sales
  2  FROM LenskartOrder;

TOTAL_SALES
-----------
       7400
```

5.5 Count Orders by Status

```
SQL> SELECT Order_Status, COUNT(*) AS Total_Orders
  2  FROM LenskartOrder
  3  GROUP BY Order_Status;

ORDER_STATUS                   TOTAL_ORDERS
------------------------------ ------------
Confirmed                                 2
Shipped                                   1
Delivered                                 1
Processing                                1
```

5.6 Calculate Average Product Price

```
SQL> SELECT ROUND(AVG(Price), 2) AS Average_Price
  2  FROM LenskartProduct;

AVERAGE_PRICE
-------------
         1480
```

5.7 Identify the Highest-Priced Product

```
SQL> SELECT Product_Name, Price
  2  FROM LenskartProduct
  3  WHERE Price = (SELECT MAX(Price) FROM LenskartProduct);

PRODUCT_NAME                                                
```

5.9 Calculate Total Amount by Order Status

```
SQL> -- 9. Total Revenue by Order Status
SQL> SELECT Order_Status, SUM(Total_Amount) AS Total_Amount
  2  FROM LenskartOrder
  3  GROUP BY Order_Status
  4  ORDER BY Total_Amount DESC;
```

```
ORDER_STATUS                   TOTAL_AMOUNT
------------------------------ ------------
Confirmed                              3200
Shipped                                1800
Delivered                              1500
Processing                              900
```